V. Conclusion

This paper has introduced the concept of iterative adaptive control and has formulated an iterative adaptive control algorithm. The controller iterations must satisfy certain convergence and continuity assumptions for the closed-loop system stability, but only one-step controller iteration is required for each new parameter update. This results in a considerable reduction in real-time computational load. The algorithm has been presented in a general form, and then applied to an adaptive LQ regulator. Based on the result of this paper, one may further investigate properties of the iterative control system when the plant is subject to disturbances, unmodeled dynamics, or time-varying parameters.

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On Adaptive Inverse Dynamics Control of Rigid Robots

MARK W. SPONG AND ROMEO ORTEGA

Abstract—**The first globally convergent adaptive inverse dynamics control law for rigid robot manipulators was derived by Craig *et al.* [4]. This scheme relies on two restrictive assumptions which are desirable to eliminate. The first is that the joint acceleration is measurable and the second is that the inverse of the estimated inertia matrix remains bounded. The need to measure joint acceleration can be overcome by the introduction of a filter as in Middleton and Goodwin [11]. The second assumption is the more restrictive, and requires modification of the parameter update law to achieve. In this short note we present an alternative formulation of adaptive inverse dynamics control which overcomes the second assumption on boundedness of the inverse of the estimated inertia matrix and the subsequent need to modify the parameter update law.**

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I. Introduction

Recently, several globally convergent adaptive control results for rigid robot manipulators have appeared [4], [14], [15]. These results, which are surveyed in [12], fall into two distinct groups. The first group is an adaptive version of the familiar inverse dynamics or computed torque control scheme which attempts to linearize and decouple the equations of motion of the robot globally. The second group relies fundamentally on the passivity of rigid robot dynamics and does not attempt to linearize the equations of motion. As in any control scheme there are advantages, disadvantages, and trade-offs in these control schemes. The passivity based control approaches are very attractive in that they do not require measurement of acceleration nor invertibility of the estimated inertia matrix. The trade-off incurred with this approach is that, even in the ideal case of perfect knowledge of the manipulator parameters, the closed-loop system remains nonlinear and coupled. Hence, the closed-loop tracking performance is more difficult to quantify. The inverse dynamics scheme of Craig *et al.* [4] leads, in the ideal case of perfect knowledge of manipulator parameters, to a closed-loop system which is linear and decoupled, which means that the closed-loop performance is easily quantified. To achieve this, however, the results of [4] require measurement of the joint acceleration and also modification of the parameter update law to guarantee invertibility of the estimated inertia matrix. Recently, the requirement that the joint acceleration be measurable was removed by the introduction of a first-order filter in [11]. The purpose of this note is to present an alternate derivation of adaptive inverse dynamics control for rigid robots that does not require invertibility of the estimated inertia matrix, and hence does not require modification of the update law as in [4].

II. Dynamics of Rigid Robots

A standard method for deriving the dynamic equations of mechanical systems is via the so-called Euler–Lagrange equations (see Spong and Vidyasagar [17] for details). Using this method the equations of motion of an n-link robot are of the form

$$\sum_j d_{kj}(q)\ddot{q}_j + \sum_{i,j} c_{ijk}(q)\dot{q}_i\dot{q}_j + \phi_k(q) = \tau_k, \qquad k = 1, \cdots, n \tag{2.1}$$

where d_{kj} are the coefficients of the inertia matrix $D(q)$, ϕ_k are gravitational forces and torques. The coefficients c_{ijk} of the Coriolis and centrifugal terms are defined as

$$c_{ijk} := \frac{1}{2}\left\{\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right\} \tag{2.2}$$

and are known as **Christoffel symbols** (of the first kind). It is common to write (2.1) in matrix form as

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau \tag{2.3}$$

where the k, jth element of the matrix $C(q, \dot{q})$ is defined as

$$\begin{aligned} c_{kj} &= \sum_{i=l}^{n} c_{ijk}(q)\dot{q}_i \\ &= \sum_{i=l}^{n} \frac{1}{2}\left\{\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right\}\dot{q}_i \end{aligned} \tag{2.4}$$

and the kth component of g is ϕ_k.

Although the equations of motion (2.1) are complex nonlinear equations for all but the simplest robots, they have several fundamental properties which can be exploited to facilitate control system design. We state these properties as follows.

Property 1: The inertia matrix $D(q)$ is symmetric, positive definite, and both $D(q)$ and $D(q)^{-1}$ are uniformly bounded as a function of $q \in \mathbb{R}^n$.

Property 2: There is an independent control input for each degree of freedom.

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Property 3: All of the constant parameters of interest such as link masses, moments of inertias, etc., appear as coefficients of known functions of the generalized coordinates. By defining each coefficient as a separate parameter, a linear relationship results so that we may write the dynamic equations (2.13) as

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Y(q, \dot{q}, \ddot{q})\theta = \tau \tag{2.5}$$

where $Y(q, \dot{q}, \ddot{q})$ is an $n \times r$ matrix of known functions, known as the **regressor**, and θ is an r-dimensional vector of parameters (see An *et al.* [2] and Khosla and Kanade [10]).

Example 2.1—Planar Elbow Manipulator: Consider the planar manipulator with two revolute joints shown in Fig. 1. Let us fix notation as follows. For $i = 1, 2$, q_i denotes the joint angle, which also serves as a generalized coordinate; m_i denotes the mass of link i, l_i denotes the length of link i; l_{ci} denotes the distance from the previous joint to the center of mass of link i; and I_i denotes the moment of inertia of link i about an axis coming out of the page, passing through the center of mass of link i. The Euler–Lagrange equations for this system can be shown to be (Spong and Vidyasagar [17])

$$d_{11}\ddot{q}_1 + d_{12}\ddot{q}_2 + c_{121}\dot{q}_1\dot{q}_2 + c_{211}\dot{q}_2\dot{q}_1 + c_{221}\dot{q}_2^2 + \phi_1 = \tau_1, \tag{2.6}$$

$$d_{21}\ddot{q}_1 + d_{22}\ddot{q}_2 + c_{112}\dot{q}_1^2 + \phi_2 = \tau_2 \tag{2.7}$$

where

$$d_{11} = m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2 + 2l_1 l_{c2}\cos q_2) + I_1 + I_2, \tag{2.8}$$

$$d_{12} = d_{21} = m_2(l_{c2}^2 + l_1 l_{c2}\cos q_2) + I_2,$$

$$d_{22} = m_2 l_{c2}^2 + I_2 \tag{2.9}$$

$$c_{121} = c_{211} = c_{221} = -c_{112} = -m_2 l_1 l_{c2}\sin q_2 := h \tag{2.10}$$

$$\phi_1 = (m_1 l_{c1} + m_2 l_1)g\cos q_1 + m_2 l_{c2} g\cos(q_1 + q_2) \tag{2.11}$$

$$\phi_2 = m_2 l_{c2} g\cos(q_1 + q_2). \tag{2.12}$$

Now if we define parameters $\theta_1, \cdots, \theta_9$ as

$$\begin{aligned} \theta_1 &= m_1 l_{c_1}^2 & \theta_4 &= m_2 l_1 l_{c_2} & \theta_7 &= m_1 l_{c_1} g \\ \theta_2 &= m_2 l_1^2 & \theta_5 &= I_1 & \theta_8 &= m_2 l_1 g \\ \theta_3 &= m_2 l_{c_2}^2 & \theta_6 &= I_2 & \theta_9 &= m_2 l_{c_2} g \end{aligned} \tag{2.13}$$

we can write (2.31), (2.32) as

$$Y(q, \dot{q}, \ddot{q})\theta = \tau \tag{2.14}$$

where $\theta = [\theta_1, \cdots, \theta_9]^T$, and Y is given by

$$Y(q, \dot{q}, \ddot{q}) = \begin{bmatrix} \ddot{q}_1 & \ddot{q}_2 & \ddot{q}_1 + \ddot{q}_2 & 2\cos q_2 \ddot{q}_1 + \cos q_2 \ddot{q}_2 - 2\sin q_2 \dot{q}_1 \dot{q}_2 - \sin q_2 \dot{q}_2^2 & \ddot{q}_1 & \ddot{q}_1 + \ddot{q}_2 & \cos q_1 & \cos q_1 & \cos(q_1 + q_2) \\ 0 & 0 & \ddot{q}_1 + \ddot{q}_2 & \cos q_2 \ddot{q}_1 + \sin q_2 \dot{q}_1^2 & \ddot{q}_2 & \ddot{q}_2 & 0 & 0 & \cos(q_1 + q_2) \end{bmatrix}. \tag{2.15}$$

Note that the choice of parameters in the above representation is not unique and that the dimension of the parameter space may depend on the particular choice of parameters. Also, not all of the parameters in the system may be unknown or it may be desired to estimate only a subset of the parameters. In this case we can write (2.14) as

$$Y_1(q, \dot{q}, \ddot{q})\theta_n + Y_2(q, \dot{q}, \ddot{q})\theta = \tau \tag{2.16}$$

where θ_n contains only known parameters and θ contains those parameters which are to be estimated. The control schemes to follow may be modified to account for this case, although, in the interest of simplicity of the exposition, we will consider only the parameterization (2.14).

III. Inverse Dynamics Based Control

In this section we summarize the inverse dynamics control idea in the known parameter case, detail the approach of Craig *et al.* [4] for adaptive inverse dynamics, and present our new scheme for overcoming the requirement that the parameter update law be modified to guarantee that the estimated inertia matrix remains invertible.

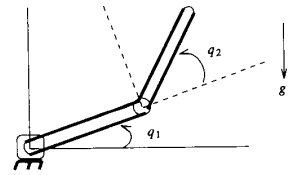

Fig. 1.

A. Known Parameter Case

The basic idea of inverse dynamics control, which is by now well known, is to seek a nonlinear control law to cancel exactly all of the nonlinear terms in the system (2.3) so that, in the ideal case, the closed-loop system is linear and decoupled. Inspecting (2.3) we see that if the control τ is chosen as

$$\tau = D(q)a + C(q, \dot{q})\dot{q} + g(q) \tag{3.1}$$

then, by substituting (3.1) into (2.3) one obtains

$$D(q)(\ddot{q} - a) = 0 \tag{3.2}$$

and Property 1 implies that

$$\ddot{q} = a. \tag{3.3}$$

The term a has the interpretation of an outer loop control law with units of acceleration, which is typically chosen as

$$a = \ddot{q}^d - K_v\dot{e} - K_p e \tag{3.4}$$

with $e = q - q^d$, where $q^d(t)$ is an n-dimensional vector of desired joint trajectories. Substituting (3.4) into (3.3) leads to the linear error equation

$$\ddot{e} + K_v\dot{e} + K_p e = 0. \tag{3.5}$$

If the gain matrices K_v and K_p are chosen as diagonal matrices with positive diagonal elements, then the closed-loop system is linear, decoupled, and exponentially stable. Global stability for this scheme is thus obvious. In fact, the closed-loop damping ratio and natural frequency may be arbitrarily assigned.

B. Adaptive Case

The adaptive implementation of the inverse dynamics control law proposed in Craig *et al.* [4] is obtained by replacing D, C, and g by their estimates, i.e.,

$$\tau = \hat{D}(q)(\ddot{q}^d - K_v\dot{e} - K_p e) + \hat{C}(q, \dot{q})\dot{q} + \hat{g}(q). \tag{3.6}$$

We assume that $\hat{D}$, $\hat{C}$, and $\hat{g}$ have the same functional form as D, C, g with estimated parameters $\hat{\theta}_1, \cdots, \hat{\theta}_r$. Thus,

$$\hat{D}\ddot{q} + \hat{C}\dot{q} + \hat{g} = Y(q, \dot{q}, \ddot{q})\hat{\theta} \tag{3.7}$$

where $\hat{\theta}$ is the vector of estimated parameters.

Substituting (3.6) into (2.3) gives

$$D\ddot{q} + C\dot{q} + g = \hat{D}(\ddot{q}^d - K_v\dot{e} - K_p e) + \hat{C} + \hat{g}. \tag{3.8}$$

Adding and subtracting $\hat{D}\ddot{q}$ on the left-hand side of (3.8) and using (3.7) we can write

$$\begin{aligned} \hat{D}(\ddot{e} + K_v\dot{e} + K_p e) &= \tilde{D}\ddot{q} + \tilde{C}\dot{q} + \tilde{g} \\ &= Y(q, \dot{q}, \ddot{q})\tilde{\theta} \end{aligned} \tag{3.9}$$

where $\tilde{(\cdot)} := (\cdot) - \hat{(\cdot)}$. Finally, the error dynamics may be written as

$$\ddot{e} + K_v\dot{e} + K_p e = \hat{D}^{-1}Y\tilde{\theta} := \Phi\tilde{\theta}. \tag{3.10}$$

We may write the system (3.10) in state space as

$$\dot{x} = Ax + B\Phi\tilde{\theta} \tag{3.11}$$

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where A is the Hurwitz matrix

$$A = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix}; \text{ and } B = \begin{bmatrix} 0 \\ I \end{bmatrix}; x = \begin{bmatrix} e \\ \dot{e} \end{bmatrix}. \tag{3.12}$$

In Craig *et al.* [4] it is assumed that $\ddot{q}$ is measurable and that $\hat{D}^{-1}$ is bounded. Under these assumptions we have the following.

Theorem 3.1: Choose the update law

$$\dot{\tilde{\theta}} = -\Gamma^{-1}\Phi^T B^T P x \tag{3.13}$$

where $\Gamma = \Gamma^T > 0$ and P is the unique symmetric positive definite solution to the Lyapunov equation

$$A^T P + PA + Q = 0 \tag{3.16}$$

for a given symmetric, positive definite Q. Under these conditions then, the solution x of (3.11) satisfies

$$x \to 0 \qquad \text{as } T \to \infty \tag{3.15}$$

with all signals remaining bounded.

Proof: Choose the Lyapunov function candidate

$$V = x^T P x + \tilde{\theta}^T \Gamma \tilde{\theta}. \tag{3.16}$$

The time derivative of V along trajectories of (3.11) is computed to be

$$\dot{V} = -x^T Q x + 2\tilde{\theta}^T [\Phi^T B^T P x + \Gamma \dot{\tilde{\theta}}]. \tag{3.17}$$

Using the parameter update law (3.11), this reduces to

$$\dot{V} = -x^T Q x \le 0. \tag{3.18}$$

This shows that $x \in L_2^{2n} \cap L_\infty^{2n}$ and $\tilde{\theta} \in L_\infty^r$. From this we conclude from (3.6) that $\tau \in L_\infty^n$. This in turn implies, using (2.1) and Property 1, that $\ddot{q} \in L_\infty^n$ and, hence, from (3.11) that $\dot{x} \in L_\infty^{2n}$. Since $\dot{x} \in L_\infty^{2n}$, x is uniformly continuous and the proof is completed using the implication

$$x \text{ uniformly continuous}, x \in L_2^{2n} \text{ implies } x \to 0 \qquad \text{as } t \to \infty. \tag{3.19}$$

We note that the parameter update law (3.13) does not guarantee *a priori* that $\hat{D}^{-1}$ remains bounded. Thus, the actual implementation of this scheme requires modification of (3.13) to ensure that $\hat{D}^{-1}$ is bounded. Practically speaking, this requires knowledge of a convex subset of the parameter space containing the true manipulators which also contains the estimated parameters for all time. To guarantee that the estimated parameters lie in this known region, the parameter update law (3.13) must be modified to project the parameter estimate vector back into this region whenever it leaves the region. We now show how to modify the inverse dynamics control law (3.6) to remove this assumption and the consequent need to modify the parameter update law.

C. A New Adaptive Inverse Dynamics Scheme

The main drawback of the result in the previous section is the requirement that $\hat{D}$ remain uniformly positive definite. We can remove this drawback by following the approach typically used in robust nonadaptive control (see, for example, Spong and Vidyasagar [16]). In this approach we choose an inverse dynamics control law of the form

$$\tau = D_0(q)(a + \delta a) + C_0(q, \dot{q})\dot{q} + g_0 \tag{3.20}$$

where $D_0 = D_0^T > 0$, C_0, g_0 are *a priori* estimates of D, C, g, respectively, with **fixed parameters**, a is given by (3.4), and δa is an additional outer loop control that compensates for the deviations ΔD, ΔC, Δg, where $\Delta(\cdot) = (\cdot)_0 - (\cdot)$. In the present setup we will choose δa adaptively. It is important to note that the terms D_0, C_0, g_0 in (3.20) are not updated on-line, and hence the invertibility of D_0 is not an issue.

If we now combine (3.20) with (2.3) we have an equation similar to (3.9)

$$\begin{aligned} D_0(\ddot{e} + K_v\dot{e} + K_p e - \delta a) &= \Delta D\ddot{q} + \Delta C\dot{q} + \Delta g \\ &= Y(q, \dot{q}, \ddot{q})\Delta\theta \end{aligned} \tag{3.21}$$

where the last equality is obtained using Property 3 of linearity in the parameters. Note, in (3.21), that $\Delta\theta = \theta_0 - \theta$ is a fixed vector in $\mathbb{R}^p$ and not a function of time, since the terms in (3.20) are fixed estimates. Finally, we write

$$\ddot{e} + K_v\dot{e} + K_p e = D_0^{-1} Y \Delta\theta + \delta a := \Phi_0 \Delta\theta + \delta a. \tag{3.22}$$

Choosing the control δa as

$$\delta a = -\Phi_0 \Delta\hat{\theta} \tag{3.23}$$

yields an equation identical to (3.11) with Φ replaced by Φ_0. Note that $\Delta\tilde{\theta} = \tilde{\theta}$ and that now Φ_0 is not a function of the estimated parameters since D_0 is fixed. Choosing an update law for $\Delta\hat{\theta}$ according to

$$\Delta\dot{\hat{\theta}} = -\Gamma^{-1}\Phi_0^T B^T P x \tag{3.24}$$

where P satisfies (3.14), and choosing a Lyapunov function candidate

$$V = x^T P x + \Delta\tilde{\theta}^T \Gamma \Delta\tilde{\theta} = x^T P x + \tilde{\theta}^T \Gamma \tilde{\theta} \tag{3.25}$$

a proof identical to that of Theorem 3.1 shows that $x \to 0$ as $t \to \infty$, with all signals remaining bounded.

IV. Conclusions

In this short note we have derived an adaptive inverse dynamics control law for rigid robots that overcomes the most restrictive difficulty of such schemes to date, namely the requirement that the inverse of the estimated inertia matrix remain bounded. Our control scheme still requires the joint accelerations for its implementation. In practice, one would implement this scheme by estimating the acceleration from the measured velocity. Good performance could still be expected without direct acceleration measurement assuming sufficient encoder resolution and sufficiently fast sampling. It would be interesting to investigate, at this point, the filtering idea of Middleton and Goodwin [11] in the context of our inverse dynamics scheme. At the present time the incorporation of the filter presented in [11] appears to be nontrivial.

Another practical advantage of our scheme over that of [4] is that, since the "nominal" inverse dynamics or inner loop portion of our control law is fixed, it is more amenable to the development of a dedicated interface and can be "burned" onto a chip to reduce on-line computation requirements.

It may be of interest to point out that, in a computer simulation of this result, the control law (3.20), (3.23) cannot be directly applied to the rigid model (2.5) as it results in an algebraic loop. This is because, in order to remove the requirement of [4] that the estimated inertia matrix remain bounded, our control algorithm requires the joint acceleration in the expression for the joint torque as well as in the parameter update law, while the result of [4] requires the acceleration only in the parameter update law. If the acceleration is obtained from a first-order filter applied to the velocity, as would typically be done in practice, then the problem with the algebraic loop is removed. We have simulated this result on a realistic robot model which includes actuator dynamics and joint compliance, in which case, again, there is no algebraic loop. Based on simulation results, which will be presented elsewhere, the algorithm in this note shows performance comparable to the results in [5].

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Minimax LQ Stochastic Tracking and Servo Problems

E. MOSCA, A. CASAVOLA, AND L. GIARRE'

Abstract—**It is proved that the separation property between feedback and feedforward of the two degrees of freedom LQ stochastic controller recently reported in [1] can be suitably extended, when a linear controller is postulated, to the minimax LQ stochastic tracking and servo problems. Besides the relevance of its results, the paper shows that in some problems, the Wiener and the H_∞ approaches can play a fruitful complementary role.**

I. Introduction

This paper considers minimax linear-quadratic (LQ) stochastic tracking and servo problems where the reference to be tracked by the output at time t is assumed to be known up to generic time $t+\tau$. Thus, our formulation unifies the servo problem ($\tau > 0$) and the tracking one ($\tau \leq 0$) into a single framework. Results and relevant references on the pure LQ stochastic version of the problem were recently reported in [1]. As discussed in Remark 1, the problem studied hereafter is intrinsically coupled with an H_∞ mixed sensitivity problem. In the H_∞ context, to the best of the authors' knowledge, the tracking problem has been considered only for $\tau = 0$ and with reference to a minimax problem not jointly involving tracking errors and plant disturbance effects (cf. [2] and references therein). On the contrary, the tracking or servo problem studied hereafter is formulated in terms of a quadratic cost including tracking errors and the worst possible plant disturbance.

II. Problem Formulation and Results

Consider a *set* of discrete-time multivariable stochastic linear *plants*

$$y(t) = \mathcal{P}(d)u(t) + \mathcal{Q}(d)e(t) \tag{1}$$

where $y(t) \in \mathcal{R}^p$ is the output, $u(t) \in \mathcal{R}^m$ is the input, $\{e(t)\}$ is a white sequence of random vectors in $\mathcal{R}^p$ with zero mean and unit covariance, and $\mathcal{P}(d) \in \mathcal{R}_{pm}(d)$ and $\mathcal{Q}(d) \in \mathcal{R}_{pp}(d)$ are causal *rational* transfer matrices in the unit backward shift operator d. $\mathcal{R}_{pm}(d)$ denotes the set of $p \times m$ matrices with elements in $\mathcal{R}(d)$, the set of polynomial fractions in the indeterminate d. Moreover,

$$\text{(A.1)} \begin{cases} \bullet\ \mathcal{P}(d) \text{ is strictly causal and free of unstable hidden modes.} \\ \bullet\ \mathcal{Q}(d) \text{ is any stable and stably invertible transfer matrix such that} \\ \|\mathcal{Q}\|_2^2 := \operatorname{tr}\left\{\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathcal{Q}^*(e^{j\omega})\mathcal{Q}(e^{j\omega})d\omega\right\} \leq 1. \end{cases}$$

Here, if $F(d)$ is any rational matrix, $F^*(d) := F^T(d^{-1})$, with the superscript T denoting transpose. Since, if $n(t) = \mathcal{Q}(d)e(t)$ and ϵ stands for expectation, $\|\mathcal{Q}\|_2^2 = \epsilon\{\|n(t)\|^2\}$, (1) represents all plants $y(t) = \mathcal{P}(d)u(t) + n(t)$ with output disturbance $\{n(t)\}$ having arbitrary rational power spectrum $\mathcal{Q}\mathcal{Q}^*$ and component variances whose sum is less than or equal to one.

Let $\omega(t) \in \mathcal{R}^p$ be the output *reference* variable. It is assumed that

(A.2) • $\{w(t)\}$ is a second-order stationary process uncorrelated with $\{e(t)\}$.

Let $I^t := \{y^t, u^{t-1}, w^{t+\tau}\}$, $y^t := \{y(t), y(t-1), \cdots,\}$ and $L_2^m(z)$ denote the subspace of $\mathcal{R}^m$-valued random vectors with finite covariance spanned by $\{z\}$. The problem that is addressed hereafter is to find among the *linear nonanticipative control laws*

$$u(t) = \mathcal{L}(I^t) \in L_2^m(I^t) \tag{2}$$

the ones that, in stochastic steady state, stabilize the plant (1) and solve the following *minimax linear-quadratic (LQ) stochastic tracking or servo problem*:

$$\inf_{\mathcal{L}} \sup_{\mathcal{Q}} \epsilon\{\|y(t) - w(t)\|_{\psi_y}^2 + \|u(t)\|_{\psi_u}^2\}. \tag{3}$$

In (3), $\psi_y = \psi_y^T \geq 0$, $\psi_u = \psi_u^T \geq 0$, and $\|v\|_\psi^2 := v^T\psi v$.

Remark 1: If $w(t) = 0$, problem (1)–(3) is the same as finding a stabilizing controller $u(t) = -K(d)y(t)$, $K(d) \in \mathcal{R}_{mp}(d)$ which minimizes

$$\sup_{0 \leq \omega < 2\pi} \lambda_{\max}[\Sigma^*(e^{j\omega})\psi_y\Sigma(e^{j\omega}) + \Gamma^*(e^{j\omega})\psi_u\Gamma(e^{j\omega})] \tag{4}$$

where $\Sigma := (I_p + \mathcal{P}K)^{-1}$ and $\Gamma := K\Sigma$ are, respectively, the *sensitivity matrix* and the *power transfer matrix* of the feedback system. The H_∞ *mixed sensitivity problem* (4) has been considered in a number of papers [3]–[8]. In particular, [5], [7], [8] showed how solutions can be obtained by polynomial equations. □

Write $\mathcal{P}$ in the form of a right coprime matrix fraction

$$\mathcal{P} = B_2A_2^{-1} \tag{5}$$

where $A_2 \in \mathcal{R}_{mm}[d]$ and $B_2 \in \mathcal{R}_{pm}[d]$ are polynomial matrices, $\mathcal{R}_{pm}[d]$ denoting the set of $p \times m$ matrices with elements in $\mathcal{R}[d]$, the set of polynomials in d. Further, define a right spectral factor $E(d) \in \mathcal{R}_{mm}[d]$ by the relation

$$A_2^*\psi_uA_2 + B_2^*\psi_yB_2 = E^*E. \tag{6}$$

It is assumed that

(A.3) • E is stable, i.e., E^{-1} is analytic on the closed unit disk.

E.g., (A.3) is fulfilled if $\psi_y > 0$ and $\psi_u > 0$.

Let a solution of the mixed sensitivity problem, viz. (1)–(3), with $w(t) \equiv 0$ be

$$u(t) = -\mathcal{R}^{-1}\mathcal{S}y(t) \tag{7}$$

where $\mathcal{R} \in \mathcal{R}_{mm}(d)$ and $\mathcal{S} \in \mathcal{R}_{pp}(d)$ are suitable stable causal transfer matrices. Specifically, $\mathcal{R} = EM_1$ and $\mathcal{S} = EN_1$ with M_1 and N_1 such that [9] $M_1A_2 + N_1B_2 = I_m$. Consequently, $\mathcal{R}$ and $\mathcal{S}$ satisfy the identity

$$\mathcal{R}A_2 + \mathcal{S}B_2 = E. \tag{8}$$

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